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Numerical Simulation of Added Mass Effects on a Hydrofoil in Cavitating Flow Using Acoustic Fluid–Structure Interaction

A fluid–structure interaction (FSI) system has been solved using the coupled acoustic structural finite element method (FEM) to simplify the cavitating flow conditions around a hydrofoil. The modes of vibration and the added mass effects have been numerically simulated for various flow conditions including leading edge attached partial cavitation on a two-dimensional NACA0009 hydrofoil. The hydrofoil has been first simulated surrounded by only air and by only water. Then, partial cavities with different lengths have been modeled as pure vapor fluid domains surrounded by the corresponding water and solid domains. The obtained numerical added mass coefficients and mode shapes are in good agreement with the experimental data available for the same conditions. The study confirms that the fluid added mass effect decreases with the cavitation surface ratio (CSR) and with the thickness of the cavitation sheet. Moreover, the simulations also predict slight mode shape variations due to cavitation that have also been detected in the experiments. Finally, the effects of changes in cavity location have been evaluated with the previously validated model. [DOI: 10.1115/1.4035113]

Keywords: hydrofoil, added mass, sheet cavitation, fluid–structure interaction, mode shape

1 Introduction

Fluid–structure interactions (FSIs) have received much attention for many decades. The dynamic response of a submerged structure in a dense fluid differs from its response in a vacuum due to the so-called “added mass effect” [1]. The added mass is relevant in many engineering applications such as offshore structures, dam–reservoir systems with seismic loads, hydrodynamics and naval structures, as well as with key components in hydro-power plants. The interest in cavitation flows is from the fact that many submerged bodies are affected by this phenomenon, such as pump impellers, turbine runners, injectors, marine propellers, hydrofoils, and others.

Large cavities attached to a surface add significant uncertainties to FSI phenomena. Both cavitating flows and classic FSI problems have been studied separately for years, but estimates of the added mass effects in cavitating flows are still difficult. Mixtures of water and vapor phases in macroscopic hydrodynamic cavities

create averaged properties that are difficult to quantify with the hydrodynamic conditions varying from the expected effects for pure liquid water flow [2]. Moreover, the boundary conditions in the cavitation regions are not fully understood. Many studies have analyzed the cavitation effects due to FSI for various bodies, especially propeller blades and hydrofoils [3,4].

For example, Amromin and Kovinskaya [5] analyzed the vibration of an elastic wing with an attached cavity in a perturbed flow. Young [6] performed a coupled three-dimensional simulation to analyze the time-dependent hydroelastic response of cavitating propellers, and Young and Savander [7] compared the mechanical efficiency of various surface-piercing propeller designs. Ducoin et al. [8] presented a method to simulate FSI problems for cavitating flows. Benaouicha and Astolfi [3] theoretically studied the effects of the added mass for cavitating flows for different body geometries. Fine et al. [9] numerically studied the added mass effects for supercavitating flows. Akcabay and Young [10] introduced a numerical viscous FSI modeling framework along with experimental measurements to study the coupled effects of unsteady cavitating flows and flexible hydrofoils. Akcabay et al. [11] simulated cavity-induced vibrations using combined numerical and experimental models.

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At present, the main numerical methods used for FSI problems include the FEM, added mass formulation, doubly asymptotic approximation (DAA), mixed boundary element method (BEM) and FEM, and the arbitrary Lagrangian–Eulerian formulation (ALE). More recently, the two-way FSI approach based on computing both the structural and fluid domains has been used in numerous studies, particularly as the codes are becoming more powerful and stable [4,6,7,11,12]. The two-way FSI method has been validated with good agreement against experimental data at the resonance frequencies [13,14]. Furthermore, some models have determined the equivalent hydrodynamic damping ratios for some critical vibration modes [15]. However, this approach requires much computational effort even for a simple FSI system such as a submerged cantilever hydrofoil. Moreover, the physical properties of the fluid medium and the boundary conditions affect the dynamic behavior of the structure, but, due to the complex formulation of the Navier–Stokes (NS) equations, their influence is difficult to quantify.

The FEM-based acoustic fluid approach that avoids the modeling effort and the time required for two-way FSI simulations can be easily applied to efficiently model complex structural and fluid geometries particularly at the design stage for safety assessments [16,17]. However, this method has not yet been used to estimate the added mass effects on hydrofoils with cavitation due to the difficulties of modeling the cavity properties. Therefore, this work assesses the benefits and limitations of the acoustic fluid hypothesis to analyze the FEM-based acoustic fluid approach with cavitation for a simple case and to evaluate its accuracy compared with experimental results.

This study presents numerical simulations of the modal response of a hydrofoil with attached partial cavities that are compared with the experimental results obtained by De La Torre et al. [18]. They investigated the influence of leading edge sheet cavitation and supercavitation on the added mass effects of a NACA0009 truncated hydrofoil in the test section of the EPFL high-speed cavitation tunnel. Their work presents added mass coefficients for various operating conditions with different cavity sizes and morphologies. The first objective of the current work is to develop a reliable FSI model that predicts added mass coefficients that are in good agreement with the experimental data. Then, the numerical model is used to further study the effects of various parameters such as the cavity dimensions and location on the vibration modes (natural frequencies and mode shapes) of the coupled system.

2 Numerical Model and Mesh

2.1 Multiphase Fluid Structure Interaction (FSI) Model.

FSI problems consider the structural dynamics equation along with the NS equations. The discretized structural dynamics equation is formulated based on structural elements. The NS equations were simplified to get the acoustic wave equation assuming that the fluid is compressible, which means that its density changes due to pressure variations (i.e., an acoustic medium) and that there is no actual fluid flow [19]. The acoustic wave equation [20] is

$$\begin{aligned} \nabla \left(\frac{1}{\rho_0} \nabla p \right) - \frac{1}{\rho_0 c^2} \frac{\partial^2 p}{\partial t^2} + \nabla \cdot \left[\frac{4\mu}{3\rho_0} \nabla \left(\frac{1}{\rho_0 c^2} \frac{\partial p}{\partial t} \right) \right] \\ = - \frac{\partial}{\partial t} \left(\frac{Q}{\rho_0} \right) + \nabla \cdot \left[\frac{4\mu}{3\rho_0} \nabla \left(\frac{Q}{\rho_0} \right) \right] \end{aligned} \quad (1)$$

where $c = \sqrt{K/\rho_0}$ is the speed of sound, ρ_0 is the mean density, K is the bulk modulus, μ is the dynamic viscosity, p is the acoustic pressure, Q is the mass source in the continuity equation, and t is the time.

Since the viscous dissipation is taken into account based on Stokes hypothesis, the wave equation is referred to as the lossy wave equation for the propagation of sound in fluids. The wave

equation is then reduced to the inhomogeneous Helmholtz equation

$$\begin{aligned} \nabla \left(\frac{1}{\rho_0} \nabla p \right) + \frac{\omega^2}{\rho_0 c^2} p + j\omega \nabla \cdot \left[\frac{4\mu}{3\rho_0} \nabla \left(\frac{1}{\rho_0 c^2} p \right) \right] \\ = -j\omega \left(\frac{Q}{\rho_0} \right) + \nabla \cdot \left[\frac{4\mu}{3\rho_0} \nabla \left(\frac{Q}{\rho_0} \right) \right] \end{aligned} \quad (2)$$

where ω is the angular frequency. Then, the boundary conditions correspond to two cases for an open or sound soft boundary

$$p = 0 \quad (3)$$

or a closed or sound hard boundary

$$\frac{\partial p}{\partial x} = 0 \quad (4)$$

The corresponding finite element formulation is obtained using the Galerkin procedure [21]. The discretized wave equation in matrix notation is [19,22]

$$[\mathbf{M}_F]\{\ddot{\mathbf{p}}_e\} + [\mathbf{C}_F]\{\dot{\mathbf{p}}_e\} + [\mathbf{K}_F]\{\mathbf{p}_e\} + \rho_0[\mathbf{R}]^T\{\ddot{\mathbf{u}}_{e,F}\} = \{\mathbf{F}_F\} \quad (5)$$

where $\mathbf{M}_F$, $\mathbf{C}_F$, and $\mathbf{K}_F$ are the fluid mass, damping, and stiffness matrices. $\mathbf{R}$ is a coupling matrix that represents the effective surface area associated with each node on the fluid–structure interface. $\mathbf{F}_F$ is the applied fluid pressure vector at the interface obtained by integrating the pressure over the surface. $\mathbf{p}_e$ is the nodal pressure vector. $\mathbf{u}_{e,F}$ is the nodal displacement component vectors of the acoustic fluid. Super-script “.” indicates first-order derivative and “..” indicates second-order derivative.

Equation (5) can be used in both liquid and gas domains. At the liquid–gas interface, $\Gamma_{\text{liquid-gas}}$, the acoustic pressure and the normal component of the velocity must be continuous. Thus

$$p_{\text{liquid}} + p_{\text{gas}} = 0 \quad \text{on } \Gamma_{\text{liquid-gas}} \quad (6)$$

$$\mathbf{n} \cdot \mathbf{u}_{F,\text{liquid}} - \mathbf{n} \cdot \mathbf{u}_{F,\text{gas}} = 0 \quad \text{on } \Gamma_{\text{liquid-gas}} \quad (7)$$

Then, the structural dynamic equation with fluid load can be written

$$\begin{aligned} [\mathbf{M}_S]\{\ddot{\mathbf{u}}_e\} + [\mathbf{C}_S]\{\dot{\mathbf{u}}_e\} + [\mathbf{K}_S]\{\mathbf{u}_e\} - [\mathbf{R}_{\text{liquid}}]\{\mathbf{p}_{e,\text{liquid}}\} \\ - [\mathbf{R}_{\text{gas}}]\{\mathbf{p}_{e,\text{gas}}\} = \{\mathbf{F}_S\} \end{aligned} \quad (8)$$

The coupling conditions on the interface between the acoustic fluid and the structure, $\Gamma_{\text{FSI-interface}}$, are given by

$$\mathbf{n} \cdot \boldsymbol{\sigma} + \mathbf{n} \cdot \mathbf{p} = 0 \quad \text{on } \Gamma_{\text{FSI-interface}} \quad (9)$$

$$\mathbf{n} \cdot \mathbf{u}_S - \mathbf{n} \cdot \mathbf{u}_F = 0 \quad \text{on } \Gamma_{\text{FSI-interface}} \quad (10)$$

where $\boldsymbol{\sigma}$ is the structure stress tensor and $\mathbf{n}$ is the outward normal unit vector of the fluid domain.

Both the discretized structural and lossy wave equations must be simultaneously considered in FSI problems, because pressure waves are generated by the structure displacement and the structure deformation is affected by the fluid pressure. This interaction at the mesh interface applies the acoustic pressure to the structure and the structural motion produces an effective “fluid load.” Then, modal analyses are used, for example, to model the acoustic modes of an acoustic cavity (standing waves) or to compute the mode shapes of vibro-acoustic systems. The governing finite element matrix equations of the multiphase FSI model are

$$\begin{aligned}
& \begin{bmatrix} [\mathbf{M}_S] & 0 & 0 \\ \bar{\rho}_{0,\text{liquid}}[\mathbf{R}_{\text{liquid}}]^T & [\mathbf{M}_{F,\text{liquid}}] & 0 \\ \bar{\rho}_{0,\text{gas}}[\mathbf{R}_{\text{gas}}]^T & 0 & [\mathbf{M}_{F,\text{gas}}] \end{bmatrix} \begin{Bmatrix} \{\ddot{\mathbf{u}}\} \\ \{\ddot{\mathbf{p}}_{\text{liquid}}\} \\ \{\ddot{\mathbf{p}}_{\text{gas}}\} \end{Bmatrix} + \begin{bmatrix} [\mathbf{C}_S] & 0 & 0 \\ 0 & [\mathbf{C}_{F,\text{liquid}}] & 0 \\ 0 & 0 & [\mathbf{C}_{F,\text{gas}}] \end{bmatrix} \begin{Bmatrix} \{\dot{\mathbf{u}}\} \\ \{\dot{\mathbf{p}}_{\text{liquid}}\} \\ \{\dot{\mathbf{p}}_{\text{gas}}\} \end{Bmatrix} \\
& + \begin{bmatrix} [\mathbf{K}_S] & -[\mathbf{R}_{\text{liquid}}] & -[\mathbf{R}_{\text{gas}}] \\ 0 & [\mathbf{K}_{F,\text{liquid}}] & 0 \\ 0 & 0 & [\mathbf{K}_{F,\text{gas}}] \end{bmatrix} \begin{Bmatrix} \{\mathbf{u}\} \\ \{\mathbf{p}_{\text{liquid}}\} \\ \{\mathbf{p}_{\text{gas}}\} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_S \\ \mathbf{F}_{F,\text{liquid}} \\ \mathbf{F}_{F,\text{gas}} \end{Bmatrix} \quad (11)
\end{aligned}$$

where $\mathbf{p}$ is the fluid pressure vector and $\mathbf{u}$ is the structure displacement vectors. $\mathbf{M}_S$, $\mathbf{C}_S$, and $\mathbf{K}_S$ are the structure mass, damping, and stiffness matrices, and $\mathbf{F}_S$ is the structural load vector.

Finally, the structure and fluid fields are combined into an unsymmetrical coupled FSI matrix system expressed as

$$\begin{pmatrix} \begin{bmatrix} [\mathbf{M}_S] & 0 & 0 \\ -\omega^2 \bar{\rho}_{0,\text{liquid}}[\mathbf{R}_{\text{liquid}}]^T & [\mathbf{M}_{F,\text{liquid}}] & 0 \\ \bar{\rho}_{0,\text{gas}}[\mathbf{R}_{\text{gas}}]^T & 0 & [\mathbf{M}_{F,\text{gas}}] \end{bmatrix} \\ +j\omega \begin{bmatrix} [\mathbf{C}_S] & 0 & 0 \\ 0 & [\mathbf{C}_{F,\text{liquid}}] & 0 \\ 0 & 0 & [\mathbf{C}_{F,\text{gas}}] \end{bmatrix} \end{pmatrix} + \begin{bmatrix} [\mathbf{K}_S] & -[\mathbf{R}_{\text{liquid}}] & -[\mathbf{R}_{\text{gas}}] \\ 0 & [\mathbf{K}_{F,\text{liquid}}] & 0 \\ 0 & 0 & [\mathbf{K}_{F,\text{gas}}] \end{bmatrix} \begin{Bmatrix} \{\phi\} \\ \{\mathbf{p}_{\text{liquid}}\} \\ \{\mathbf{p}_{\text{gas}}\} \end{Bmatrix} = 0 \quad (12)$$

where $\{\omega\}$ are the natural circular frequencies and $\{\phi\}$ are the modal shapes of the structure. For a linear system, $\{\mathbf{u}\} = \{\phi\} \cos \omega t$.

The relevant simulations were completed in ANSYS MECHANICAL software. Due to the incapability on this issue of the ANSYS default configuration, supplementary codes based on ANSYS APDL language were implemented.

2.2 FEM Model and Mesh Sensitivity Analysis. The FSI model was evaluated by comparing to experimental data for a NACA0009 hydrofoil for various cavitation conditions tested at the EPFL high-speed cavitation tunnel as described in Ref. [18]. This test facility is a closed loop with a rectangular test section with $150 \times 150 \times 750$ mm inner dimensions. The NACA0009 hydrofoil had a truncated trailing edge with a chord of 100 mm and a span of 150 mm. The truncated trailing edge thickness was 3.22 mm. The foil was made of an aluminum alloy whose properties are given in Table 1. The hydrofoil body was machined to insert two piezoelectric patches and the corresponding cables as shown in left of Fig. 1. A spanwise roughness strip was added on the leading edge as shown in right of Fig. 1 to trigger a more uniform and homogeneous cavitation sheet along the profile span. The strip was a 2 mm band of sand glued on the suction side next to the stagnation point with a thickness of approximately 200 μm .

The FSI domain, which included the entire test section of the cavitation tunnel, was meshed with hexahedral elements as shown in Fig. 2 with the hydrofoil in horizontal position at the center of the fluid domain.

A sensitivity analysis (left side of Fig. 2) was performed to determine an appropriate mesh. The numbers of elements in the span and chordwise directions were increased while predicting the natural frequencies using a traditional modal analysis. The left side of Fig. 3 shows the percent deviations of the three first natural frequencies (f_1 , f_2 , and f_3) as the number of elements in the spanwise direction was increased for the hydrofoil in air.

The fluid domain around the hydrofoil (right side of Fig. 2) was then modeled with acoustic elements in the tunnel test section

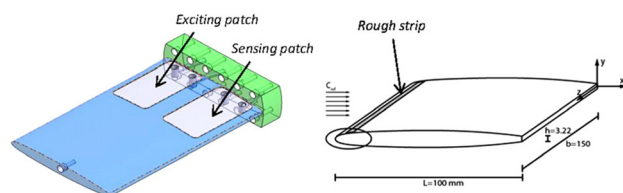


Fig. 1 Truncated NACA0009 design (left) and details of the leading edge roughness (right) (from Ref. [18])

with inner dimensions of $150 \times 150 \times 750$ mm. The hydrofoil was clamped on one side with the opposite tip separated from the tunnel wall by a small fluid gap of 0.12 mm.

The number of elements in the fluid volume was increased in all directions for the mesh sensitivity analysis. The largest mesh had 100 elements in the spanwise direction and 70 elements in the other two directions. The right side of Fig. 3 shows the percent changes of f_1 , f_2 , and f_3 as the number of elements in the vertical direction is increased for the hydrofoil in water.

The results in Fig. 3 show that the frequencies converge faster in air than in water. Therefore, the largest number of elements was used for both the fluid and structure to guarantee the mesh independence. Similarly, one unique element was eventually used to mesh the gap distance of 0.12 mm as proved to be accurate in the sensitivity tests carried out by De La Torre et al. [23].

One lateral section of the hydrofoil was constrained in the model. The rest of the hydrofoil surfaces were in contact with the

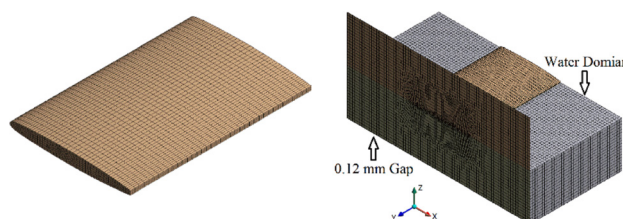


Fig. 2 Discretized finite element mesh on the NACA0009 hydrofoil (left) and in the FSI domain with the surrounding water (right)

Table 1 Properties of the hydrofoil material

Properties	Young's modulus (GPa)	Density (kg/m ³)	Poisson's ratio
Value	0.54	2770	0.33

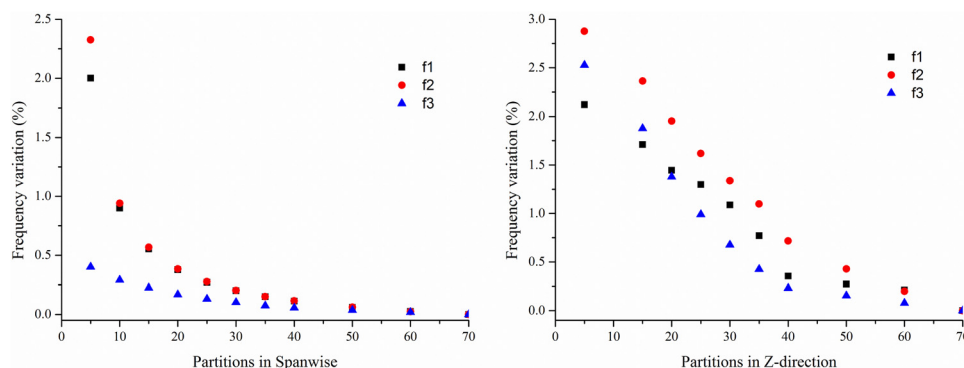


Fig. 3 Frequency variations for the first three modes as the number of elements in the spanwise direction was increased in the hydrofoil without fluid domain (left) and in the vertical direction in the fluid (right) with the complete FSI domain

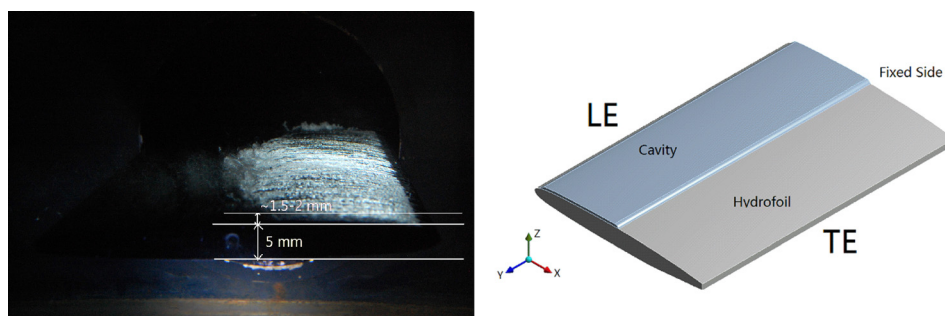


Fig. 4 Lateral view of the cavity in the cavitation tunnel (left) and corresponding model of the attached cavity (right) (left photograph from Ref. [2])

fluid. A fluid structure interaction condition was set at these interfaces so that the displacement of the structure was the same as that of the fluid in the normal direction. The fluid boundaries were modeled as fully reflective due to the coupling characteristics of vibrations in water. The water properties were a density of 1000 kg/m^3 and a speed of sound of 1450 m/s .

The attached cavities were assumed to have a constant length, l , along the whole span as shown on the right side of Fig. 4. Acoustic elements were used as for the water domain. The acoustic pressure and the normal component of the water/vapor velocity were assumed to be continuous at the interface of the vapor cavity with the surrounding water. The vapor properties were a density of 0.174 kg/m^3 and a speed of sound of 343.24 m/s .

3 Comparison of FEM Predictions and the Experimental Results

The added mass effects were evaluated by means of the added mass coefficient, C_M , defined for the three first natural frequencies of the hydrofoil which correspond to the first bending (f_1), the torsion (f_2), and the second bending (f_3) modes of vibration as

$$C_M = \left(\frac{f_{\text{vacuum } i}}{f_{\text{fluid } i}} \right)^2 - 1 \quad (13)$$

where $f_{\text{vacuum } i}$ and $f_{\text{fluid } i}$ are the natural frequencies in air and in water of mode i .

The cavity size was characterized by the CSR defined as the ratio of the area of the hydrofoil surface covered by the sheet cavity (as shown in Fig. 5) and the total hydrofoil surface including the suction and pressure sides

$$\text{CSR} = \frac{l \cdot s}{2 \cdot c \cdot s} = \frac{l}{2c} \quad (14)$$

where l is the cavity length, c is the hydrofoil chord, and s is the hydrofoil span.

The cavitation number, σ , was defined as

$$\sigma = (p_r - p_v) / (1/2 \rho V^2) \quad (15)$$

where p_r and p_v are the reference and vapor pressures of the fluid and V is the fluid velocity.

3.1 No Cavitation. The numerical and experimental results for the three first modes in air surrounded by water and with the test section filled only up to the lower hydrofoil surface (so-called half-wetted condition) are listed in Table 2. The maximum error is less than 8% which indicates an acceptable level of agreement.

3.2 Partial Cavitation. From Eq. (12), it can be seen that the main parameters that affect the simulation results (natural frequencies and mode shapes) are the cavity location, its average density, and the speed of sound inside it. So, in order to simulate any incidence angle cavitation conditions, it is only necessary to change these cavity properties (e.g., cavity location, length, and void ratio) in the acoustic FSI model to the ones estimated in the experiment.

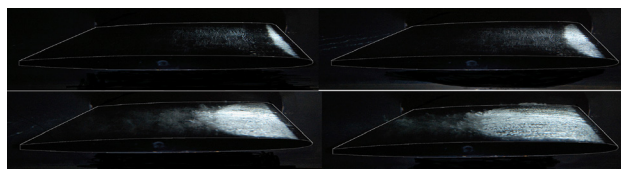


Fig. 5 Hydrodynamic conditions resulting in increasing cavitation surface ratios (photographs from Ref. [18])

Table 2 Numerical and experimental natural frequencies of the hydrofoil in air, completely submerged in still water, and partially submerged in still water (half-wetted)

	First bending f_1 (Hz)			First torsion f_2 (Hz)			Second bending f_3 (Hz)		
	Air	Water	Half-wetted	Air	Water	Half-wetted	Air	Water	Half-wetted
Sim.	283.6	134.96	170.76	990.2	606.01	721.93	1705.3	959.28	1143.2
Exp.	270.2	130.2	163	1018.6	614.8	755	1671	886	1113.6
Error (%)	4.72	3.53	4.54	-2.87	-1.45	-4.58	2.01	7.64	2.59

Table 3 Predicted C_M for various cavity thicknesses with comparisons with experimental data for $l/c = 0.318$

Thickness (mm)	C_{M2}		C_{M3}	
	Exp. Sim.	1.61 Error (%)	Exp. Sim.	1.92 Error (%)
1.0	1.662	3.26	2.043	6.38
1.5	1.652	2.62	1.953	3.06
1.8	1.617	0.43	1.932	0.63
2.0	1.598	-0.72	1.894	-1.38
2.5	1.561	-2.84	1.856	-3.33

Table 4 Numerical and experimental C_M 's for attached cavities of different lengths

l/c	σ	CSR	C_{M2}			C_{M3}		
			Exp.	Sim.	Error (%)	Exp.	Sim.	Error (%)
0.02	1.539	0.01	1.6	1.705	6.56	2.35	2.223	-5.40
0.054	1.006	0.027	1.82	1.690	-4.49	2.08	2.176	4.63
0.114	0.79	0.057	1.73	1.670	-3.48	1.99	2.126	6.86
0.318	0.6	0.159	1.61	1.617	0.43	1.92	1.932	0.63
0.442	0.543	0.221	1.26	1.550	23.03	1.81	1.786	-1.34

Based on such assumption and for simplification, the current numerical results were obtained for a 0 deg incidence angle mesh, but the cavity properties set in the model were those corresponding to an upstream velocity of 14 m/s and a 1 deg incidence angle. In this case, the experimental results uncertainty was estimated to be approximately 9% for l/c and CSR, 3% for σ , 2% for C_M , and approximately 1% for the frequencies as stated in Ref. [18].

The left side of Fig. 4 shows the side view of the hydrofoil in the test section used to approximate the thickness of the attached cavity. The photographs taken during the tests showed thicknesses ranging from around 1 to 2.5 mm.

Due to this uncertainty, a parametric study was conducted by progressively increasing the cavity thickness in the model within

this range to evaluate the effects of this uncertainty on the errors. This procedure simulated a pure vapor cavity with an l/c ratio of 0.318.

The C_M numerical results for f_2 and f_3 in Table 3 show that the added mass effect for both modes decreases with the cavity thickness. The best C_M results were obtained for a cavity thickness of 1.8 mm with less than 1% of error. Therefore, a cavity thickness of 1.8 mm was used for the rest of the calculations assuming that this dimension did not change significantly with the cavity size.

The final results for C_{M2} and C_{M3} for the CSR's from 0.01 to 0.221 are listed in Table 4. All the errors are below 7% with the exception of C_{M2} with an error around 23% for the largest CSR of 0.221. These errors might be associated to differences in both the cavity thicknesses and the average properties of the actual cavities. The plots in Fig. 6 show that the numerical and experimental C_M results present follow similar trends as a function of CSR and σ ; thus, reinforcing the validity of the simulation approach.

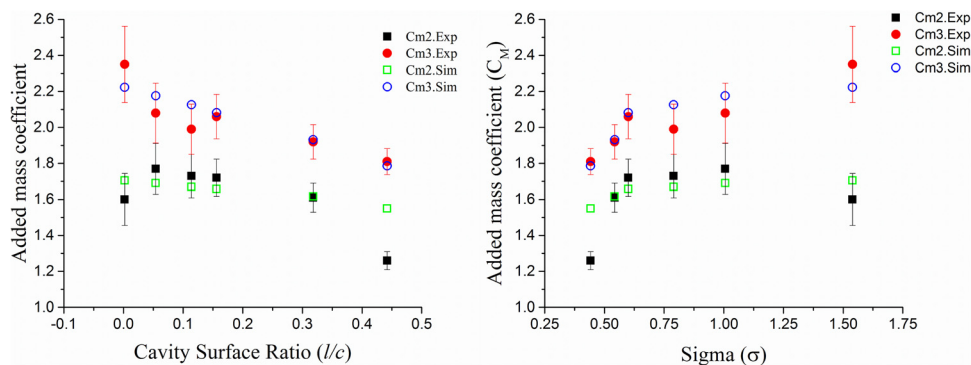
The effects of the boundary conditions on the acoustic mode shapes can be seen in Fig. 7. The relative (normalized) deformation of the hydrofoil for the three vibration modes is compared for the cases having air, still water, half-wetted condition, a short cavity, and a medium-size cavity. For the two cases with cavitation, the size and location of the vapor cavity is indicated by a square with a yellow dotted perimeter.

For f_1 , the mode shape is almost constant for any fluid condition. The maximum deformation is always found at the tip section.

For f_2 , the cases with the still fluid and those with an attached cavity are quite different. For the still fluid, the point with the maximum deformation is located in the corner between the tip section and the trailing edge. With the attached cavity, this point has moved to the corner between the tip section and the leading edge.

Finally, for f_3 , the colored deformation contours clearly show that the symmetry along the central span is lost for the two cavitation conditions. In these cases, the trailing edge presents a larger curvature than the leading edge between the fixed section and the nodal line. The nodal line breaks in two close to the trailing edge for both cavitation conditions.

The significant differences in the f_3 mode shapes shown in Fig. 7 between air and cavitation conditions obtained with the acoustic model are in close agreement with experimental measurements [24]. The mode shapes were measured using a laser

**Fig. 6 Predicted and measured added mass coefficients for various CSR and sigma for f_2 and f_3**

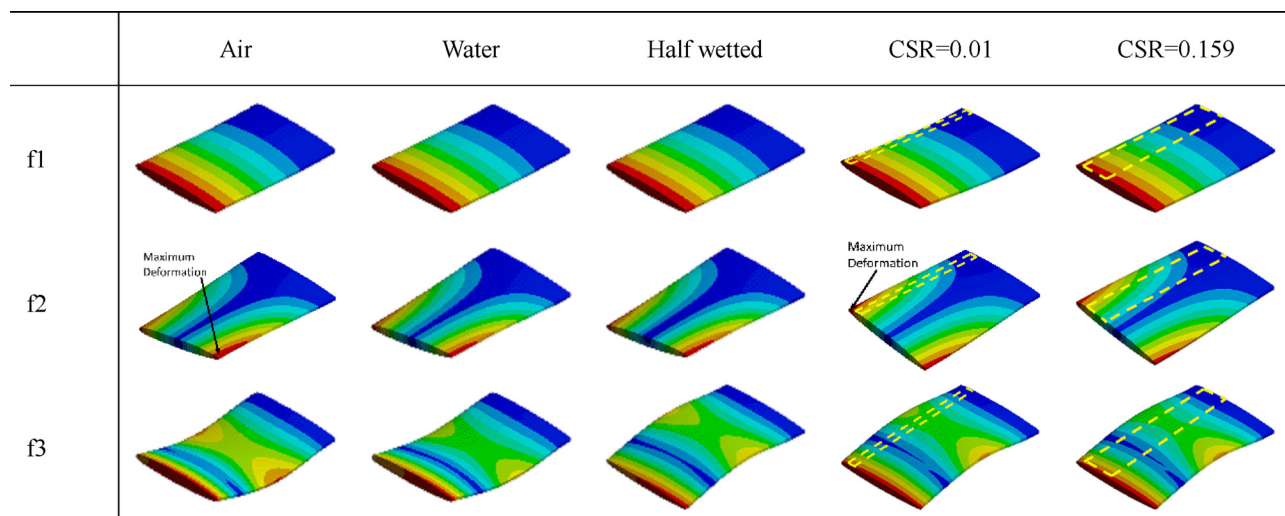


Fig. 7 Numerical hydrofoil mode shapes in air, still water, half-wetted, and with short and medium length cavities

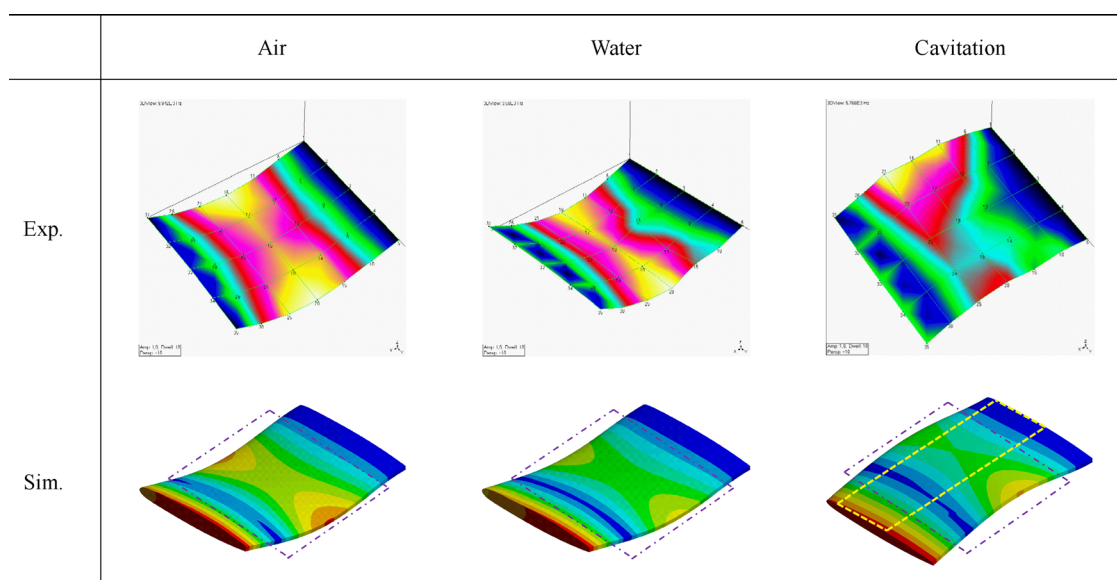


Fig. 8 Measured (top) and predicted (bottom) f_3 mode shapes in air, in water, and with an attached cavity of l/c around 0.5 (experimental results from Ref. [24])

Doppler vibrometer operating at resonance forced by embedded piezoelectric ceramic (PZT) patches. Figure 8 compares the experimental mode shapes in air, in water, and with an attached cavity of l/c around 0.5 with the simulation results. It must be noted that the experimental results do not show the whole surface of the hydrofoil in spanwise direction, because the upper transparent window of the tunnel test section was too narrow [23]. The actual visible area has been marked on the simulated mode shapes with a dotted dash line.

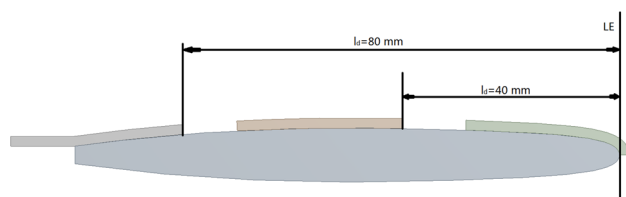


Fig. 9 Cross section of the hydrofoil for cavity lengths, l_d , of 0, 40, and 80 mm

In summary, the vibration modes most affected by the cavitation are f_2 and f_3 . These changes are induced by the local reduction of the added mass on the hydrofoil due to the presence of pure vapor instead of water. Moreover, the specific location of the vapor sheet on the hydrofoil also affects the mode shape changes.

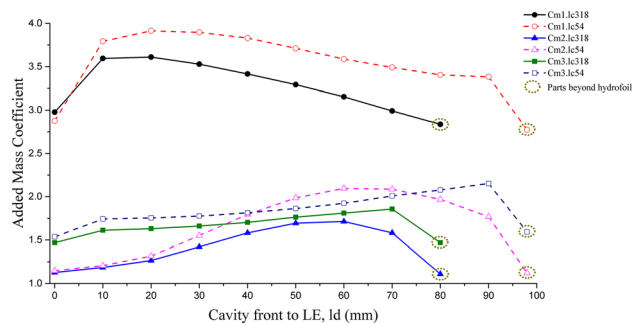


Fig. 10 Added mass coefficients for various l_d

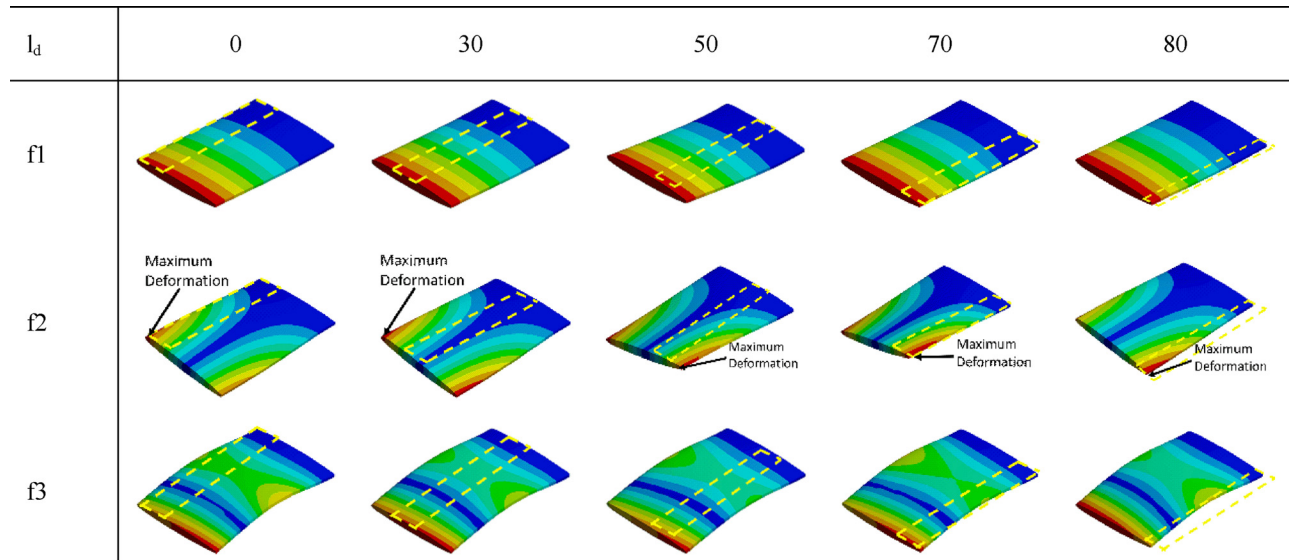


Fig. 11 Mode shapes of the hydrofoil with the cavity at different locations for $l/c = 0.318$

Therefore, the validated numerical model can now be used to simulate the effects of the cavity location.

3.3 Cavity Location. The numerical model was then used to evaluate the effects on the added mass coefficients and mode shapes of a constant length cavity starting at different distances, l_d , from the leading edge as shown in Fig. 9 for three cases corresponding to 0, 40, and 90 mm.

In Fig. 10, the added mass coefficients for two cavity lengths, $l/c = 0.318$ and 0.054 , are plotted for distances l_d ranging from 0 to 80 mm and from 0 to 98 mm, respectively, in steps of 10 mm. For all three modes, C_M decreases significantly when the cavity starts at 80 mm or 98 mm when part of the cavity is beyond the hydrofoil. On the other side, C_{M1} for f_1 increases significantly when the cavity is displaced 10 mm from the leading edge, then C_{M1} starts to decrease with increasing l_d . For f_2 , C_{M2} increases with l_d to a maximum around 60 mm and then begins to decrease. Finally for f_3 , C_{M3} keeps increasing until the cavity extends beyond the hydrofoil when it suddenly decreases as already stated. Similar hydrodynamic coefficient drops were found by Akcabay and Young [10] when the cavity extended beyond the trailing edge. Moreover, their study also showed that the flow fluctuations and the hydrofoil deformations were reduced once the cavity length exceeded the chord length and in supercavitation flows.

The effects of cavity position on the mode shapes are shown in Fig. 11. The f_1 mode shape remains almost constant for each case. For f_2 , the point with the maximum deformation changes from the leading to the trailing edge side on the tip when the cavity is located downstream along the symmetry axis of the mode. For f_3 , the mode shape is not constant along the chord as is the case without cavitation. In general, the edge of the hydrofoil that is farther from the cavity presents a larger curvature than the opposite edge when the whole cavity is attached to the hydrofoil surface. Moreover, the maximum deformation occurs at the corner between the tip section and the trailing edge. However, when the cavity extends beyond the trailing edge (i.e., $l_d = 80$ mm and 98 mm), the trailing edge presents a larger curvature than the leading edge and the maximum deformation point moves to the opposite corner between the tip section and the leading edge.

4 Conclusions

The FSI effects on the dynamic response of a NACA0009 hydrofoil were numerically simulated for various flow conditions

including an attached cavity using an acoustic FEM method. A wide range of cases was analyzed from still fluids (air and water) to attached cavities (vapor–water) with various sizes. The obtained added mass coefficients for the three first modes of vibration have maximum deviations of around 10% from measured values, and the mode shapes are also in good agreement with the actual hydrofoil deformations. Therefore, the decrease of the added mass effect with the CSR as well as the mode shape variations due to cavitation can be well predicted with the presented numerical modal analysis.

The validated numerical model shows that the added mass coefficients also decrease with the cavity thickness. Moreover, the location of where the cavity starts relative to the leading edge can modify the added mass coefficients and the mode shapes depending on the vibration mode. These effects appear to be more important for higher modes of vibration.

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References

- [1] Brennen, C., 1982, "A Review of Added Mass and Fluid Inertial Forces," Department of the Navy, Port Hueneme, CA, Report No. CR 82.010.
- [2] De La Torre, O., 2013, "Influence of Cavitation on the Dynamic Response of Hydrofoils," Ph.D. thesis, Universitat Politècnica de Catalunya, Barcelona, Spain.
- [3] Benaouicha, M., and Astolfi, J.-A., 2012, "Analysis of Added Mass in Cavitating Flow," *J. Fluids Struct.*, **31**(8), pp. 30–48.
- [4] Chae, E. J., Akcabay, D. T., and Young, Y. L., 2013, "Dynamic Response and Stability of a Flapping Foil in a Dense and Viscous Fluid," *Phys. Fluids (1994-Present)*, **25**(10), p. 104106.
- [5] Amromin, E., and Kovinskaya, S., 2000, "Vibration of Cavitating Elastic Wing in a Periodically Perturbed Flow: Excitation of Subharmonics," *J. Fluids Struct.*, **14**(5), pp. 735–751.
- [6] Young, Y., 2007, "Time-Dependent Hydroelastic Analysis of Cavitating Propulsors," *J. Fluids Struct.*, **23**(2), pp. 269–295.
- [7] Young, Y. L., and Savander, B. R., 2011, "Numerical Analysis of Large-Scale Surface-Piercing Propellers," *Ocean Eng.*, **38**(13), pp. 1368–1381.
- [8] Ducoin, A., Young, Y. L., and Sigrist, J.-F. O., 2010, "Hydroelastic Responses of a Flexible Hydrofoil in Turbulent, Cavitating Flow," *ASME Paper No. FEDSM-ICNMM2010-30310*.
- [9] Fine, N. E., Uhlman, J. S., and Kring, D. C., 2001, "Calculation of the Added Mass and Damping Forces on Supercavitating Bodies," California Institute of Technology, Pasadena, CA, accessed July 1, 2016, <http://resolver.caltech.edu/CAV2001:sessionB3.006>
- [10] Akcabay, D. T., and Young, Y. L., 2014, "Influence of Cavitation on the Hydroelastic Stability of Hydrofoils," *J. Fluids Struct.*, **49**(35), pp. 170–185.

- [11] Akcabay, D. T., Chae, E. J., Young, Y. L., Ducoin, A., and Astolfi, J. A., 2014, "Cavity Induced Vibration of Flexible Hydrofoils," *J. Fluids Struct.*, **49**(35), pp. 463–484.
- [12] Wu, Q., Huang, B., Wang, G., and Gao, Y., 2015, "Experimental and Numerical Investigation of Hydroelastic Response of a Flexible Hydrofoil in Cavitating Flow," *Int. J. Multiphase Flow*, **74**, pp. 19–33.
- [13] Dompierre, F., and Sabourin, M., 2010, "Determination of Turbine Runner Dynamic Behaviour Under Operating Condition by a Two-Way Staggered Fluid-Structure Interaction Method," *IOP Conf. Ser.: Earth Environ. Sci.*, **12**(1), p. 012085.
- [14] Hübner, B., Seidel, U., and Roth, S., 2010, "Application of Fluid-Structure Coupling to Predict the Dynamic Behavior of Turbine Components," *IOP Conference Series: Earth and Environmental Science*, IOP Publishing, Timișoara, Romania, p. 012009.
- [15] Liu, X., Luo, Y., Karney, B., Wang, Z., and Zhai, L., 2015, "Virtual Testing for Modal and Damping Ratio Identification of Submerged Structures Using the PolyMAX Algorithm With Two-Way Fluid-Structure Interactions," *J. Fluids Struct.*, **54**, pp. 548–565.
- [16] Presas, A., Valero, C., Huang, X., Egusquiza, E., Farhat, M., and Avellan, F., 2012, "Analysis of the Dynamic Response of Pump-Turbine Runners-Part I: Experiment," *IOP Conference Series: Earth and Environmental Science*, IOP Publishing, Beijing, China, p. 052015.
- [17] He, L., Wang, Z., Kurosawa, S., and Nakahara, Y., 2014, "Resonance Investigation of Pump-Turbine During Startup Process," *IOP Conference Series: Earth and Environmental Science*, IOP Publishing, Montreal, Canada, p. 032024.
- [18] De La Torre, O., Escaler, X., Egusquiza, E., and Farhat, M., 2013, "Experimental Investigation of Added Mass Effects on a Hydrofoil Under Cavitation Conditions," *J. Fluids Struct.*, **39**(5), pp. 173–187.
- [19] ANSYS, 2015, "ANSYS Theory Reference," ANSYS, Canonsburg, PA.
- [20] Junger, M. C., and Feit, D., 1986, *Sound, Structures, and Their Interaction*, MIT Press, Cambridge, MA.
- [21] Bathe, K. J., 2006, *Finite Element Procedures*, Prentice Hall, Upper Saddle River, NJ.
- [22] Zienkiewicz, O. C., and Newton, R. E., 1969, "Coupled Vibrations of a Structure Submerged in a Compressible Fluid," *Symposium on Finite Element Techniques*, Stuttgart, Germany, pp. 360–378.
- [23] De La Torre, O., Escaler, X., Egusquiza, E., and Farhat, M., 2014, "Numerical and Experimental Study of a Nearby Solid Boundary and Partial Submergence Effects on Hydrofoil Added Mass," *Comput. Fluids*, **91**(7), pp. 1–9.
- [24] De La Torre, O., Escaler, X., Egusquiza, E., and Farhat, M., 2015, "Experimental Mode Shape Determination of a Cantilevered Hydrofoil Under Different Flow Conditions," *Proc. Inst. Mech. Eng., Part C*, epub.